

1. TALK 1. DEFINITION OF AN ELEMENTARY TOPOS (IVAN)

In this talk I defined an elementary topos and tried to motivate the definition.

Definition 1.1. An *elementary topos* is a category $\mathcal{E}$ such that:

- $\mathcal{E}$ has all finite limits;
- $\mathcal{E}$ is cartesian closed, i.e. for any object A , the functor $- \times A : \mathcal{E} \rightarrow \mathcal{E}$ has a right adjoint, denoted by $(-)^A : \mathcal{E} \rightarrow \mathcal{E}$;
- $\mathcal{E}$ has a subobject classifier, i.e. there exists an object Ω and a morphism $t : 1 \rightarrow \Omega$ such that for any (equivalence class of a) monomorphism $m : A' \hookrightarrow A$, there exists a unique morphism $\chi_m : A \rightarrow \Omega$ making the following diagram a pullback:

$$\begin{array}{ccc} A' & \longrightarrow & 1 \\ m \downarrow & & \downarrow t \\ A & \xrightarrow{\chi_m} & \Omega \end{array}$$

Slogan: *A topos is a generalized space*¹.

1.1. finite limits. I want to think about my category $\mathcal{E}$ as a "space" and objects of $\mathcal{E}$ as its small "subspaces".

Thus, I want to be able to define points of my small subspaces, and to be able to take intersections of them.

To define generalized points, we need to have a terminal object 1 in $\mathcal{E}$.

Definition 1.2. A *global element* of an object A is a morphism $1 \rightarrow A$.

To define intersections, we need to have pullbacks in $\mathcal{E}$. Whenever we have two "subspaces" U and V of a "space" X , we can define their intersection as the pullback:

$$\begin{array}{ccc} U \times_X V & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

Remark 1.3. In particular, it implies existence of finite products (take X to be the terminal object) and equalizers (take two maps from U to V and X to be $V \times V$).

As a consequence of required terminal object and pullbacks, we obtain that $\mathcal{E}$ has all finite limits:

Theorem 1.4. *Let $\mathcal{E}$ be a category with a terminal object and pullbacks. Then $\mathcal{E}$ has all finite limits.*

1.2. subobject classifier. This part is the most difficult for me to motivate by means of geometry. At the end I will try to do that, but for now let us consider a prototypical example:

Example 1.5. Consider the category of sets **Set** and an arbitrary set X .

Then, there are two ways to characterize any given subset $S \subseteq X$:

- as a monomorphism $S \hookrightarrow X$;
- as a characteristic function $\chi_S : X \rightarrow \{0, 1\}$, where $\chi_S(x) = 1$ if $x \in S$ and $\chi_S(x) = 0$ otherwise. So that the subset S is the preimage of 1 under χ_S .

In fact, such a property serves as a core duality between substructures of a given object X and certain functions out of it. One might recognize this idea in many fields starting from algebraic geometry and ending with (maybe) logic.

¹For more slogans, see <https://ncatlab.org/nlab/show/topos>

So, this property is very desirable to have in our category $\mathcal{E}$, and it is rather geometric than one might think at first glance.

Definition 1.6. A *subobject classifier* in a category $\mathcal{E}$ is an object Ω together with a morphism $t : 1 \rightarrow \Omega$ such that for any (equivalence class of a) monomorphism $m : A' \hookrightarrow A$, there exists a unique morphism $\chi_m : A \rightarrow \Omega$ making the following diagram a pullback:

$$\begin{array}{ccc} A' & \longrightarrow & 1 \\ m \downarrow & & \downarrow t \\ A & \xrightarrow{\chi_m} & \Omega \end{array}$$

Remark 1.7. Equivalently, the functor that sends an object A to the set of subobjects of A is representable:

$$\begin{array}{ccc} \text{Hom}(X, \Omega) & \xrightarrow{\sim} & \{\text{equiv. classes of mono } S \hookrightarrow X\} \\ \parallel & & \\ \text{Sub}_{\mathcal{E}} X & \longleftarrow & \text{This is a functor via pullbacks} \end{array}$$

Example 1.8. As we have already seen, in the category of sets **Set**, the subobject classifier is the set $\{0, 1\}$ and the morphism $t : 1 \rightarrow \{0, 1\}$ sends the unique element of 1 to 1 .

Example 1.9. A slightly more interesting example is the category of presheaves $\mathbf{Set}^{\mathcal{C}^{op}}$ on a small category $\mathcal{C}$.

We want to construct a presheaf Ω such that:

$$\text{Hom}_{\text{PSh}(\mathcal{C})}(h_C, \Omega) \cong \text{Sub}(h_C)$$

By Yoneda lemma, the left hand side is isomorphic to $\Omega(C)$. Thus, we need to construct a presheaf Ω such that $\Omega(C)$ is the set of subobjects of h_C for any object $C \in \mathcal{C}$.

$$\Omega(c) := \text{Sub}_{\text{PSh}}(h_c)$$

A morphism from a terminal object to Ω is a morphism from the terminal presheaf to Ω . The terminal presheaf is the constant presheaf that sends any object of $\mathcal{C}$ to the singleton set.

Thus, we need to construct a morphism $t : 1 \rightarrow \Omega$ which picks a subfunctor of h_C for any object $C \in \mathcal{C}$; it is given by picking h_C itself.

Definition 1.10. A subfunctor of a representable functor h_C is called a *sieve* on C .

A more explicit description of the sieve on C is the following: it is a collection of morphisms with codomain C that is closed under precomposition, i.e. if $f : D \rightarrow C$ is in the sieve and $g : E \rightarrow D$ is an arbitrary morphism, then $f \circ g : E \rightarrow C$ is also in the sieve.

1.3. cartesian closedness. Recall that we tend to work with objects of our topos as with small patches. In particular, we want to have the "mapping space" object of the same nature. However, a hom-set doesn't belong to our topos in a canonical way.

Definition 1.11. An *exponential object* is a right adjoint to the functor $- \times A : \mathcal{E} \rightarrow \mathcal{E}$. We denote it by $(-)^A : \mathcal{E} \rightarrow \mathcal{E}$.

Definition 1.12. A category $\mathcal{C}$ is called *cartesian closed* if it has a terminal object, finite products, and exponentials.

Proposition 1.13. For any small category $\mathcal{C}$, the presheaf category $\mathbf{Set}^{\mathcal{C}^{op}}$ is cartesian closed.

Proof. (sketch) Let P, Q be two presheaves on $\mathcal{C}$ and assume that the exponential object Q^P exists. Then, by the definition of exponential object, we have the following natural isomorphism:

$$Q^P C \cong \text{Hom}_{\text{PSh}}(h_C, Q^P) \cong \text{Hom}_{\text{PSh}}(h_C \times P, Q)$$

So, let us define the exponential object Q^P by the formula:

$$Q^P C := \text{Hom}_{\text{PSh}}(h_C \times P, Q)$$

□

Side remark 1.14. Does this remind you of $\text{map}_{\text{sSet}}(X, Y)_n = \text{Hom}_{\text{sSet}}(X \times \Delta^n, Y)$?

2. TALK 2. BOOLEAN AND HEYTING ALGEBRAS (KASRA)

As I promised at the beginning, I will heavily rely on a categorical perspective while summarizing the content of the talks.

Definition 2.1. • A *preorder* is a category in which any hom-set is either empty or a singleton.
• A *poset* is a preorder in which there are no isomorphisms except identities.

Remark 2.2. Any skeleton of a preorder is a poset.

Example 2.3. The closed interval $[0, 1]$ with the usual order is a bounded lattice: the meet of two elements is their minimum, the join is their maximum, and the bottom and top elements are 0 and 1.

Definition 2.4. • A *lattice* is a poset that has all finite limits and colimits. They are called meet (denoted by $\wedge$) and join (denoted by $\vee$), respectively.
• A *distributive lattice* is a lattice in which the following distributive law holds:

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

Equivalently, products and coproducts distribute over each other.

Remark 2.5. In some literature, authors distinguish between the terms "lattice" and "bounded lattice", requiring the existence of an initial and a terminal object in the latter case, that is, colimits and limits over the empty diagram.

Example 2.6. The real line $\mathbb{R}$ with the usual order is an unbounded lattice: meets and joins are given by minimum and maximum, but there is no least or greatest element.

Example 2.7. For any topological space X , the collection of open subsets $\mathcal{O}(X)$ ordered by inclusion is a bounded lattice: the meet of two opens is their intersection, the join is their union, the bottom element is the empty set $\emptyset$, and the top element is X .

Definition 2.8. A *complement* of an element x in a (bounded) lattice is an element y such that $x \wedge y = 0$ and $x \vee y = 1$.

Definition 2.9. A *Boolean algebra* is a distributive (bounded) lattice in which every element has a complement.

Definition 2.10. A *Heyting algebra* is a bounded lattice which is cartesian closed. An exponential object is usually denoted by $x \Rightarrow y$ and called an implication.

Thus, we have:

$$z \leq (x \Rightarrow y) \Leftrightarrow z \wedge x \leq y$$

In particular, we have $y \leq (x \Rightarrow y)$, which can be thought of as a map $Y \rightarrow \text{Map}(X, Y)$.

Example 2.11. The collection of open subsets $\mathcal{O}(X)$ of a topological space X is a Heyting algebra, where the implication $U \Rightarrow V$ is given by the interior of the union of the complement of U and V .

Proposition 2.12. *Any Boolean algebra is a Heyting algebra.*

Proof. The implication $x \Rightarrow y$ is given by the formula $\neg x \vee y$. □

Example 2.13. The collection of open subsets $\mathcal{O}(X)$ of a topological space X is not a Boolean algebra, since the complement of an open set is not necessarily open.

Some useful properties of Heyting algebras:

- Units and counits of the adjunction become:

$$x \leq (y \Rightarrow (x \wedge y)) \quad \text{and} \quad y \wedge (y \Rightarrow x) \leq x$$

- The properties of exponentials become:

$$(x \Rightarrow 1) = 1 \quad \text{and} \quad (1 \Rightarrow x) = x$$

- The functor $x \Rightarrow (-)$ is a right adjoint, so it preserves $\wedge$ and the functor $(- \wedge y)$ is a left adjoint, so it preserves $\vee$.

Remark 2.14. The latter property implies that the underlying lattice of a Heyting algebra is distributive:

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

For any Heyting algebra, we can define a negation operator, which is in some sense similar to the complement in a Boolean algebra, but **it is not a complement!**

Definition 2.15. The *negation* of an element x in a Heyting algebra is defined as $\neg x := (x \Rightarrow 0)$.

From a logical perspective, the negation can be described as "if x then false", which is not always the same as "not x ". This crucial distinction is a point of a subject called constructive mathematics. I personally recommend Bauer's paper *Five stages of accepting constructive mathematics*, [1]. I would be happy to discuss it in the future if there is interest.

Proposition 2.16. *In a Heyting algebra, if an element x has a complement, that complement must be $\neg x$.*

In fact, this is an example of one of the most general and important phenomena in mathematics, which brings together in a broad sense all sorts of dualities in mathematics.

Side remark 2.17. The keyword here is a **dualizable object**; see, as always, the relevant *nLab* page².

- In the case of closed categories, we have a certain kind of internal dual, with examples like:
Vector spaces (and its dual given by a hom to the base field)³,
Heyting algebras (with an element and its negation, "mapping space" to 0).
Pontryagin duality (duality between certain topological groups and their character groups, with the dualizable object being S^1)⁴
- Algebra/Geometry paradigm
Gelfand duality (duality between commutative C^* algebras and compact Hausdorff spaces, with the dualizable object being $\mathbf{C}$)⁵,
Stone duality (duality between Boolean algebras and Stone spaces, with the dualizable object being $\{0, 1\}$)⁶ and many more.

²See <https://ncatlab.org/nlab/show/dualizable+object>.

³See <https://ncatlab.org/nlab/show/dual+vector+space>.

⁴See <https://ncatlab.org/nlab/show/Pontryagin+dual>.

⁵See <https://ncatlab.org/nlab/show/Gelfand+duality>.

⁶See <https://ncatlab.org/nlab/show/Stone+duality>.

Let's return to the Heyting algebras.

Proposition 2.18. *In any Heyting algebra H ,*

- $x \leq \neg\neg x$, $x \leq y$ implies $\neg y \leq \neg x$
- $\neg x = \neg\neg\neg x$
- $\neg\neg(x \wedge y) = \neg\neg x \wedge \neg\neg y$
- $(x \Rightarrow x) = 1$
- $x \wedge (x \Rightarrow y) = x \wedge y$, $y \wedge (x \Rightarrow y) = y$
- $x \Rightarrow (y \wedge z) = (x \Rightarrow y) \wedge (x \Rightarrow z)$

Conversely, in any lattice L with 0 and 1 a binary operation $\Rightarrow$ satisfying these identities must be the implication of a Heyting algebra structure on the lattice L .

Remark 2.19. Try to find examples of similar properties in other categories.

Example 2.20 (Negation in the Heyting Algebra $\text{Open}(X)$). Let $\text{Open}(X)$ be the Heyting algebra of all open subsets of a topological space X .

In any Heyting algebra, the implication operation $\Rightarrow$ is defined as the right adjoint to the meet operation, satisfying the condition $z \leq (x \Rightarrow y)$ if and only if $z \wedge x \leq y$. The negation $\neg x$ is fundamentally defined as $x \Rightarrow 0$ (implying falsity). By substituting $y = 0$, this adjunction yields:

$$(9) \quad z \leq \neg x \quad \text{iff} \quad z \wedge x = 0$$

Because $\neg x$ is the largest element satisfying this property, it follows that in a complete lattice, the negation $\neg x$ can be explicitly constructed as the supremum (or union) of all those elements z which meet x in 0:

$$\neg x = \bigvee \{z \mid z \wedge x = 0\}$$

Translating this algebraic property to the spatial context of $\text{Open}(X)$, the negation $\neg U$ of an open subset $U \subseteq X$ is the union of all open subsets of X which do not meet U . Consequently:

- $\neg U$ is the interior of the set-theoretic complement of U :

$$\neg U = \text{Int}(X \setminus U)$$

- Alternatively, because the complement of the closure equals the interior of the complement, this can be expressed as:

$$\neg U = X \setminus \text{Cl}(U)$$

From this definition, the double negation $\neg\neg U$ becomes the interior of the closure of U :

$$\neg\neg U = \text{Int}(\text{Cl}(U))$$

which may be strictly larger than U itself. Thus, this example demonstrates that $\neg\neg x$ need not equal x .

Moreover, $\neg x$ is not necessarily a complement of x ; though $x \wedge \neg x = 0$ always holds, it may not be the case that $x \vee \neg x = 1$.

There is a certain interplay between the notions of complement (which is for Boolean algebras) and negation (which is for Heyting algebras).

Proposition 2.21. *In a Heyting algebra, if an element x has a complement, that complement must be $\neg x$.*

It turns out that all elements of a Heyting algebra have complements if they are "finite" in some sense.

Proposition 2.22. *A Heyting algebra is a Boolean algebra if and only if $\neg\neg x = x$ for all x , or, if and only if $\neg x \vee x = 1$ for all x .*

Side remark 2.23. *I intentionally mentioned the "finite" nature of Boolean algebras. In many examples, the certain notions of finiteness are defined by the isomorphism of an object with its double dual. I don't know the conceptual reason for this, and would be glad to learn if you know it.*

The following theorem should probably have been stated earlier, as it provides a motivation for the whole talk. Why are we interested in Boolean and Heyting algebras?

Theorem 2.24. *Consider the category of presheaves $PSh(\mathcal{C})$ on a small category $\mathcal{C}$. Then, for any presheaf $P \in PSh(\mathcal{C})$, its subobjects form a Heyting algebra.*

Remark 2.25. The same is true for any topos.

Example 2.26. In the particular case of the category of G -sets $= \text{Fun}(\text{BG}, \text{Sets})$, a subobject of a given G -set X is a subset that is closed under the G -action. The negation is given simply by taking the complement of the subset.

3. TALK 3. SHEAVES, SIEVES AND EXAMPLES. (WILLIAM)

REFERENCES

- [1] ANDREJ BAUER *Five stages of accepting constructive mathematics*. Bulletin (New Series) of the American Mathematical Society, 54(3):481–498, 2017.